

Algorithms and criteria for use in syphilis diagnostics

Persons with clinical symptoms and/or signs consistent with late syphilis (tertiary, visceral, especially involving the cardiovascular system, musculoskeletal system, etc.) should be examined using a non-treponemal test, as well as two treponemal tests: ELISA or FTA, and TPHA or TPI.

Persons without clinical signs of disease but who have had sexual exposure or close household contact with individuals suffering from contagious forms of syphilis should be tested as suspected cases of primary syphilis.

Persons without signs of disease but suspected of latent syphilis must be tested using a non-treponemal test and two treponemal tests. Acceptable combinations include: ELISA or FTA + TPI; ELISA or FTA + TPHA.

Neonates born to mothers seropositive for syphilis who did not receive adequate therapy or were untreated during pregnancy must be tested using a non-treponemal test and tests for early infection — FTA, or, if available, 19S-IgM FTA-ABS or ELISA-IgM.

Neonates born to mothers seropositive for syphilis who received adequate treatment should be tested using MPR (or RPR). If the result is positive, further testing with ELISA for whole antibodies or ELISA-IgG, FTA, and, if possible, 19S-IgM FTA-ABS or ELISA-IgM (if available) should be performed.

Children suspected of late congenital syphilis should be tested using the MPR test and two treponemal tests: ELISA, FTA, TPHA, or TPI.

Diagnostic Principles

It is essential to remember that the only reliable laboratory criterion for diagnosing active syphilis is the direct detection of *Treponema pallidum* via

dark-field microscopy, direct immunofluorescence, or PCR from lesions typical of early syphilis.

Serological methods, when clinical symptoms are present, can confirm the diagnosis with high certainty.

Serological results should be interpreted cautiously in the absence of clinical signs or symptoms.

Initial Testing Algorithm

For patients presenting with complaints or signs suggestive of syphilis, or individuals tested due to contact with a confirmed case:

- If clinical signs suggest early syphilis, perform direct detection of *T. pallidum* using dark-field microscopy, direct immunofluorescence, or PCR.
- If positive: diagnosis of syphilis is confirmed.
- If negative: repeat the test. Proceed with serological testing using an appropriate scheme.
- Perform serological testing:
 - If the screening test (non-treponemal or treponemal) is positive, conduct confirmatory serology (treponemal or non-treponemal, as appropriate).
 - Refer the patient to a specialist in dermatology and venereal diseases.

Interpretation of Serological Tests

Result of serologic testing	Conclusion
Non-treponemal test positive, Treponemal test negative	False-positive non-treponemal screening test
Non-treponemal test positive, Treponemal test positive	Untreated syphilis; previously treated late syphilis
Non-treponemal test negative, Treponemal test positive	Very early untreated syphilis; previously treated early syphilis
Non-treponemal test negative, Treponemal test negative	Not syphilis; incubating syphilis; very late syphilis; syphilis with concomitant HIV infection and immunosuppression

Latent Syphilis

If latent syphilis is suspected, the condition is likely to be identified through serological testing. A quantitative non-treponemal test should be performed, and the patient referred to a specialist for treatment and ongoing monitoring.